The SS Guardian

A Remote-Controlled Airboat for Plastic Debris Collection in Central Texas Waterways

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Abstract & Background

A $200 remote-controlled airboat that reaches shoreline trash where standard boats can't. Built to extend Austin Watershed Protection's manual cleanup, not replace it.

Lady Bird Lake and Lake Austin collect plastic from urban runoff, storm drains, and creeks. Recent UT Jackson School research shows microplastic concentrations in lake sediments are rising [3]. Most microplastic begins as larger plastic that breaks down over time [1]. Intercepting macroplastic upstream prevents future microplastic [2].

Cleanup today is conducted through manual labor by watershed crews. The hardest spots to reach, shallow vegetated shorelines under bridges and brush, are exactly where debris piles up.

Project Objectives

Build a small, low-cost watercraft that:

- Reaches shallow, vegetated zones other boats can't
- Collects plastic without harming wildlife
- Remains under $250 with accessible materials
- Deployable by two people in under two hours
- Extends volunteer cleanup reach

Design Methods

Design Process

Stakeholder interviews → House of Quality → engineering specs. Five concepts scored via Pugh charts; airboat won on maneuverability, vegetation tolerance, and cost.

		Concepts				
		Drone (Datum)	Net Gun	Floating Channel	Creek Crawler	Air Boat
Criteria	Plastic Removal Effectiveness	0	0	+	-	+
	Wildlife Safety	0	+	-	+	0
	Public Transparency	0	-	+	+	0
	Collection Capacity	0	-	+	+	+
	Ease of Maintenence	0	+	-	0	+
	Ease of Deployment Relocation	0	+	-	+	+
	Durability and Reliability	0	+	0	+	+
	Noise and Visual Intrusion	0	+	0	0	0
	Cost and Ease of Manufacturing	0	+	+	0	+
	Mobility	0	-	-	-	-
	Risk of Technical Development	0	+	0	0	0
Totals	Positive Count	0+	7	4	5	6
	Negative Count	0-	3	4	2	1
	Total Sum	0	4	0	3	5
	Rank	X	2	4	3	1

Figure 1. Pugh concept-selection matrix. Five design candidates were scored against stakeholder-weighted criteria; the airboat led on maneuverability, vegetation tolerance, and cost.

Final Design & System Arch.

Hull

24" XPS foam core with fiberglass + epoxy skin; 1" draft, 3 kg loaded.

Propulsion

Above-water 10×7 nylon prop in a 3D-printed shroud — no submerged drivetrain.

Debris capture

One-piece TPU rake across the bow; flexes around brush without snagging.

Steering

Servo-driven air rudder in the prop wash; responds down to <0.5 m/s.

Electronics

7.2V pack, ESC, 2.4 GHz RC receiver; single servo, no tether.

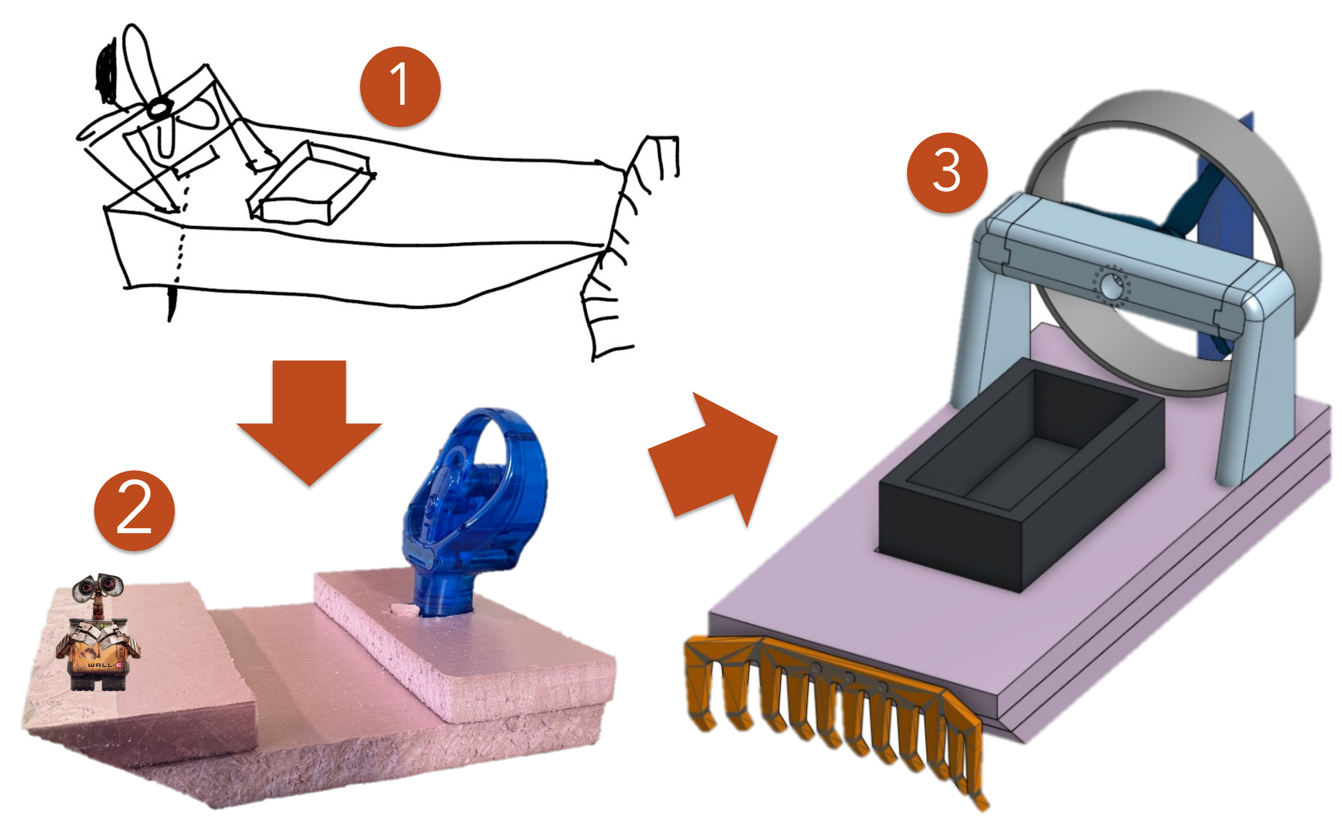

Figure 2. Design evolution: (1) initial sketch, (2) foam-and-fan proof of concept (Fall 2025), (3) final CAD (Spring 2026).

Manufacturing & Testing

Materials

Foam-safe epoxy and XPS-compatible glue for the hull; waterway-safe paint. 3D-printed parts in PLA (shroud, spoiler, rudder, servo mount) and TPU (rake).

Test-driven iterations

- **Motor mount.** PLA spoiler split and the press-fit prop slipped → 3-piece tabbed spoiler with a collet-mounted nylon prop.
- **Rake.** Two-piece PLA snapped at the seam → one-piece TPU that flexes around brush.
- **Rudder.** Water rudder bent the servo arm → upsized air rudder with corrected print orientation.

Results & Conclusion

Performance

3 kg loaded · 1.2 m/s top · 1" draft · 20–30 min · $200

$F_{thrust} = \rho \cdot A \cdot v_{exit}^2 = (1.2\ kg/m^3)(0.051\ m^2)(20\ m/s)^2 \approx 24\ N$ vs. hull drag at 1 m/s ≈ 7.5 N → ~3× thrust-to-drag margin.

Honest limits

Small payload. Requires an operator. A complement to industrial interceptors like Mr. Trash Wheel [4] and autonomous skimmers like WasteShark [5], not a substitute.

What's next

Brushless electronics · fine-mesh shroud for low-hanging brush · twin rudders on a second servo · longer hull for greater payload · stiffer rake for heavier loads · higher-capacity battery

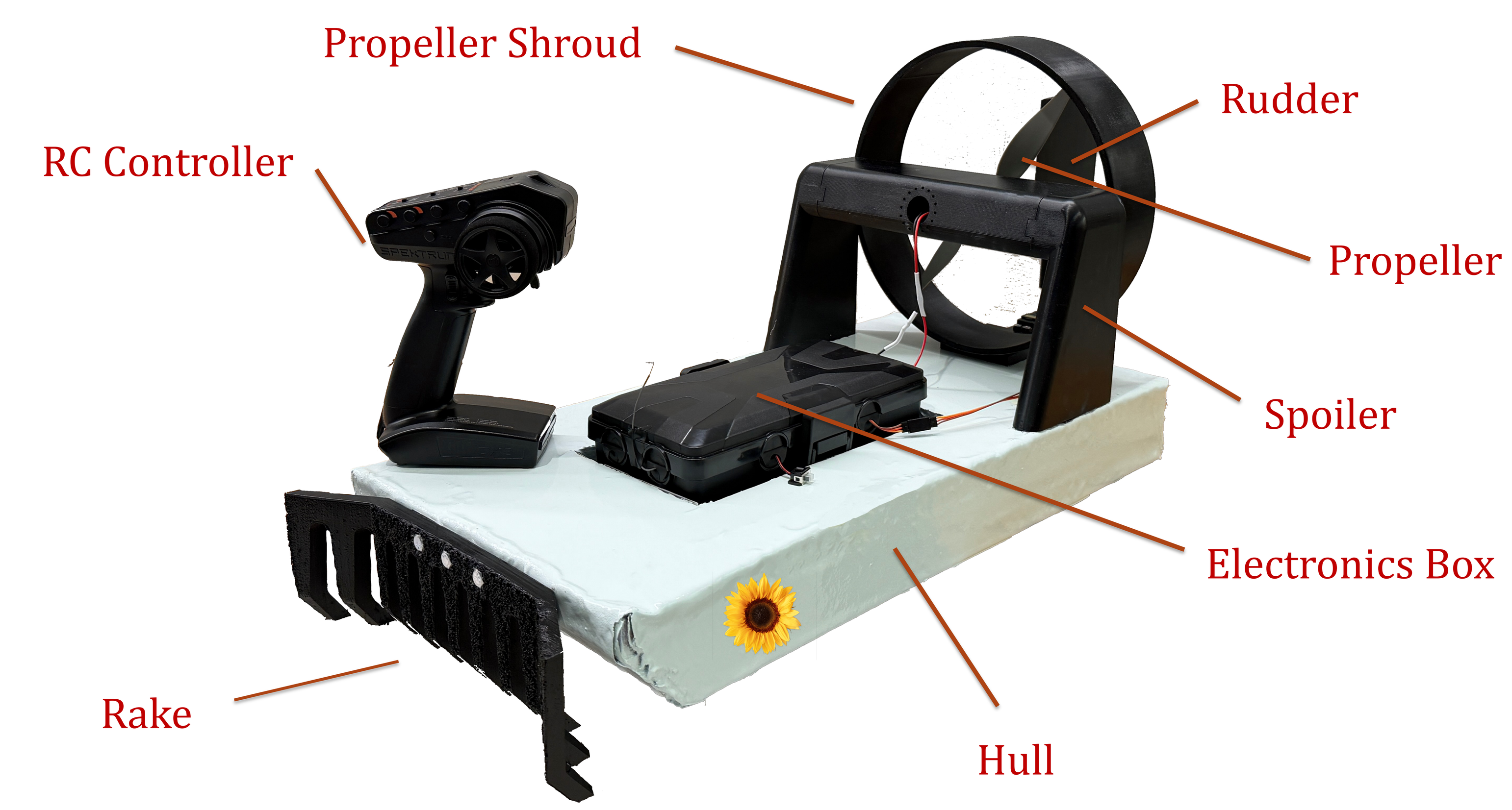

Figure 3. The SS Guardian, final build, with part callouts.

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